

### Meets the requirements of ASTM D 4434, Type III

#### Features and Components

**Advanced Solid Phase Polymer Formulation:** Using the optimal amount of Elvaloy® KEE (Ketone Ethylene Ester) polymer to: ensure plasticizer retention, extend roof life (*exceeded 40,000 hours of accelerated weathering testing - ASTM G 154 requires 5,000 hours*), and to reduce maintenance costs.

**Patented Aramid-Reinforced Edge:** Aramid fiber is woven into the fastening side of PVC membrane.

**Spunbond 3.8 oz. Polyester Fleece Back Mat:** Interlocking, multiple-layer, uniformly arranged continuous filament strands are needle punched with thousands of barbed needles, creating an extremely durable, strong yet light and flexible protection layer.

**Non-wicking Reinforced Polyester Scrim:** Our fully integrated manufacturing process adds tensile strength and toughness. Due to the non-wicking edge sealant is not required.

**Excellent Chemical Resistance:** JM PVC is inherently resistant to oils, air conditioning coolants, fuels and grease.

**Energy Savings:** The White, Grey ES and Sandstone ES provide exceptional reflectivity and emissivity for energy savings.



Component

**M**  
Membrane

Single Ply

Type

**FB**  
Fleece Back

#### Colors\*

| Grey  | Grey ES | Sandstone | Sandstone ES |
|-------|---------|-----------|--------------|
| White |         |           |              |

\* All colors not available as standard stocked items in all size configurations. Please call for minimums and lead times.

**System Compatibility** This product may be used as a component in the following systems. Please reference product application for specific installation methods and information.

| Multi-Ply | BUR                                                  |    |    | APP |    |    | SBS |    |  |  |  |
|-----------|------------------------------------------------------|----|----|-----|----|----|-----|----|--|--|--|
|           | HA                                                   | CA | HW | HA  | CA | HW | SA  | MF |  |  |  |
|           | Compatible with the selected multi-ply systems above |    |    |     |    |    |     |    |  |  |  |

| Single Ply | TPO                                                   |    |    |    | PVC |    |    | EPDM |    |    |
|------------|-------------------------------------------------------|----|----|----|-----|----|----|------|----|----|
|            | MF                                                    | AD | SA | IW | MF  | AD | IW | MF   | AD | BA |
|            | Compatible with the selected single ply systems above |    |    |    |     |    |    |      |    |    |

Key: HA = Hot Applied CA = Cold Applied HW = Heat Weldable SA = Self Adhered MF = Mechanically Fastened IW = Induction Weld BA = Ballasted AD = Adhered

\*Can be used as a cap sheet in BUR and SBS systems when adhered using hot asphalt.

#### Energy and the Environment

| Standard            |                 |            | Reflectivity | Emissivity |
|---------------------|-----------------|------------|--------------|------------|
| CRRC®               | White           | Initial    | 0.86         | 0.86       |
|                     |                 | 3 Yr. Aged | 0.70         | 0.82       |
|                     | Sandstone ES    | Initial    | 0.73         | 0.83       |
|                     |                 | 3 Yr. Aged | 0.58         | 0.82       |
|                     | Grey ES         | Initial    | 0.67         | 0.85       |
|                     |                 | 3 Yr. Aged | 0.54         | 0.82       |
| CA Title 24         | White           | Pass       | 0.86         | 0.86       |
| LEED®<br>(SRI)      | White           | Initial    | 108          |            |
|                     |                 | 3 Yr. Aged | 84           |            |
|                     | Sandstone ES    | Initial    | 89           |            |
|                     |                 | 3 Yr. Aged | 67           |            |
|                     | Grey ES         | Initial    | 80           |            |
|                     |                 | 3 Yr. Aged | 61           |            |
| Recycled<br>Content | Post-consumer   |            | 0%           |            |
|                     | Post-industrial |            | 0% - 10%     |            |

The LEED® Solar Reflectance Index (SRI) is calculated per ASTM E1980.

#### Peak Advantage® Guarantee Information

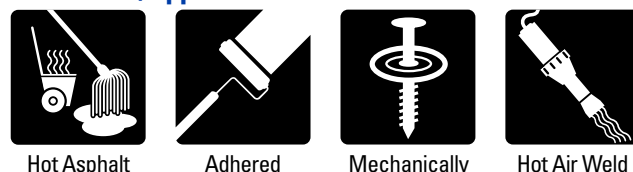
Enhanced guarantees are now available on certain systems for wind and puncture. Consult your local sales representative for more information and for specific guarantee terms and costs.

| Product                            | Guarantee Term |
|------------------------------------|----------------|
| When used in most JM PVC Systems** | Up to 30 years |

\* Elvaloy® is a registered trademark of Dow.

\*\*Contact JM Technical Services for specific systems.

#### Installation/Application



Refer to JM PVC application guides and detail drawings for instructions.

Hot asphalt application with 12' FB requires two hot installers to ensure the asphalt stays warm enough to receive the fleece.

#### Packaging and Dimensions

| Sizes                                     | Coverage                        |               |
|-------------------------------------------|---------------------------------|---------------|
| 6.33' x 75' (1.93 m x 22.86 m)            | 474.75 ft² (44.11 m²)           |               |
| 12' x 75' (3.66 m x 22.86 m) (white only) | 900 ft² (83.61 m²) (white only) |               |
| Widths                                    | 6.33'                           | 12'           |
| Rolls per Pallet                          | 10                              | 7             |
| Pallet Weight - lb (kg)                   | 2740 (1242.8)                   | 3843 (1743.2) |
| Pallets per Truck*                        | 14                              | 8             |
| Producing Locations                       | Pawtucket, RI and Lancaster, SC |               |

\*Assumes 48' flatbed truck and does not reflect pallets of accessories or impact of mixed sizes.

#### Codes and Approvals



### Meets the requirements of ASTM D 4434, Type III

#### Tested Physical Properties

| Physical Properties   |                                                         | ASTM Test Method            | ASTM Requirements    | JM PVC FB – 80 mil               |
|-----------------------|---------------------------------------------------------|-----------------------------|----------------------|----------------------------------|
| Strength              | Breaking Strength, min, lbf (N)                         | D 751                       | 200 (890)            | 511 (2,273)                      |
|                       | Elongation at Break, min %                              | D 751                       | 15                   | 42                               |
|                       | Tearing Strength, min, lbf (N)                          | D 751                       | 45 (200)             | 84.6 (376)                       |
|                       | Seam Strength, min, % of breaking strength              | D 751                       | 75                   | 93                               |
|                       | Static Puncture Resistance, lbf (kg)                    | D 5602                      | Pass @ 33 (15)       | Pass                             |
|                       | Dynamic Puncture Resistance, J                          | D 5635                      | Pass @ 20            | Pass                             |
| Longevity             | Thickness, min, in.                                     | D 751                       | +/- 10% from Nominal | 0.080 (Nominal)                  |
|                       | Thickness Over Scrim, min, in.                          | D 7635                      | 0.016                | 0.038                            |
|                       | Water Absorption, max, %                                | D 570 modified              | 3.0                  | 0.41                             |
|                       | Low Temperature Bend, °F                                | D 2136                      | No Cracks @ -40°F    | Pass                             |
| Heat Aged Performance | Properties after Heat Aging, min                        | D 3045                      | 56 days @ 176°F      |                                  |
|                       | Breaking Strength, % (after aging)                      | D 751                       | 90                   | 92                               |
|                       | Elongation, % (after aging)                             | D 751                       | 90                   | 94                               |
|                       | Linear Dimensional Change, max, % (after 6 hrs @ 176°F) | D 1204                      | 0.5                  | 0.2                              |
| Weather Performance   | Accelerated Weathering, min                             | G 151 & G 154               | 5,000 hrs            |                                  |
|                       | Cracking (@ 7x magnification)                           | G 154                       | No Cracks            | Pass @ >40,000 hrs               |
|                       | Discoloration (by observation)                          | G 154                       | Negligible           | Negligible                       |
|                       | Crazing (@ 7x magnification)                            | G 154                       | No Crazing           | Pass @ >40,000 hrs               |
|                       | Moisture Vapor Transmission                             | ASTM E 96, Proc B, Method A |                      | 0.01 g/m <sup>2</sup> per 24 hrs |

| Physical Properties         |  | FM Test Method | FM Requirements | JM PVC FB – 80 mil |
|-----------------------------|--|----------------|-----------------|--------------------|
| Foot Traffic Test, lbf (kg) |  | FM 4470        | 200 (90.7)      | Pass               |

Technical specifications as shown in this literature are intended to be used as general guidelines only. Please refer to the Safety Data Sheet and product label prior to using this product. The Safety Data Sheet is available by calling (800) 922-5922 or on the web at [www.jm.com/roofing](http://www.jm.com/roofing). The physical and chemical properties of the product listed herein represent typical, average values obtained in accordance with accepted test methods and are subject to normal manufacturing variations. They are supplied as a technical service and are subject to change without notice. Check with the regional sales representative nearest you for current information.

All Johns Manville products are sold subject to Johns Manville's standard Terms and Conditions, which includes a Limited Warranty and Limitation of Remedy. For a copy of the Johns Manville standard Terms and Conditions or for information on other Johns Manville roofing products and systems, visit [www.jm.com/terms-conditions](http://www.jm.com/terms-conditions).